5. CSS — UI Styling & Interactions

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These rules style the leave form, the buttons, and the September payslip. Motion sits inside a reduced-motion check. `@keyframes` is named at the end and is not used in the labs.

74. Styling HTML Forms

Labels stack above the fields. Fields share a width, a border, and padding. The leave form stays in normal flow under the directory. Set `box-sizing: border-box` on the universal selector so the padding stays inside the width. UI component boundaries need about 3:1 contrast against the adjacent surface. A light grey like `#d0d7de` on white is only about 1.5:1 — use a darker line such as `#6e7781`.

Example

```
/* styles.css */
* {
  box-sizing: border-box;
}

label {
  display: block;
  margin-bottom: 0.25rem;
}

input, select, textarea {
  width: 100%;
  max-width: 24rem;
  padding: 0.5rem;
  border: 1px solid #6e7781;
  font: inherit;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Leave</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <h2>Leave request</h2>
    <form action="/leave" method="post">
      <p>
        <label for="reason">Reason</label>
        <textarea id="reason" name="reason" rows="4"></textarea>
      </p>
      <button type="submit">Submit leave</button>
```

```
    </form>
  </main>
</body>
</html>
```

This will produce the following result:



75. Styling Buttons

A button and a link that acts as a button share one class. A link styled as a button uses `inline-flex`. The label then lines up in the middle of the padding. Colour is not the only sign of the control. The text says what it does.

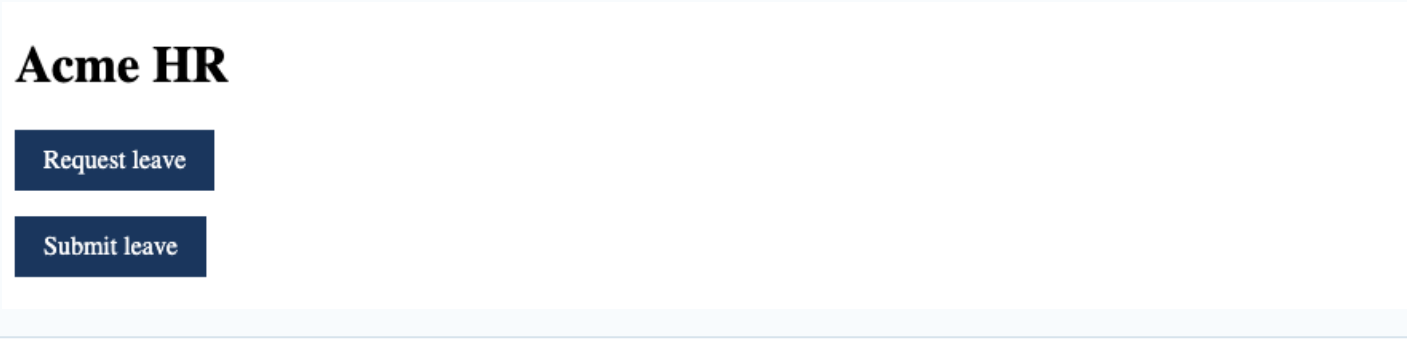
Example

```
/* styles.css */
button, a.button {
  display: inline-flex;
  align-items: center;
  padding: 0.5rem 1rem;
  border: 2px solid #1a365d;
  background-color: #1a365d;
  color: #ffffff;
  font: inherit;
  text-decoration: none;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Buttons</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <p><a class="button" href="leave.html">Request leave</a></p>
    <form action="/leave" method="post">
      <button type="submit">Submit leave</button>
```

```
    </form>
  </main>
</body>
</html>
```

This will produce the following result:



76. Styling Tables

The payslip uses collapsed borders, padding in the cells, and right-aligned amounts. The header row has a light navy surface. The footer row carries the net pay. Numbers line up when the amount column is right-aligned.

Example

```
/* styles.css */
table {
  border-collapse: collapse;
  width: 100%;
  max-width: 32rem;
}

th, td {
  border: 1px solid #6e7781;
  padding: 0.5rem;
  text-align: left;
}

thead th {
  background-color: #e8eef6;
}

td {
  text-align: right;
}

th[scope="row"] {
  text-align: left;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Payslip</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
```

```
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <h2>Payslip</h2>
    <table>
      <caption>Payslip for Ananya Rao, September 2026</caption>
      <thead>
        <tr>
          <th scope="col">Component</th>
          <th scope="col">Amount</th>
        </tr>
      </thead>
      <tbody>
        <tr>
          <th scope="row">Basic</th>
          <td>₹50,000</td>
        </tr>
        <tr>
          <th scope="row">HRA</th>
          <td>₹20,000</td>
        </tr>
        <tr>
          <th scope="row">Net pay</th>
          <td>₹78,800</td>
        </tr>
      </tbody>
    </table>
  </main>
</body>
</html>
```

This will produce the following result:

Acme HR

Payslip

Payslip for Ananya Rao, September 2026

Component	Amount
Basic	₹50,000
HRA	₹20,000
Net pay	₹78,800

77. Cards & Navigation Bars

The nav is a row of links in the header. Each employee card has a border, padding, and the status badge from the positioning section. The header uses Flexbox. The cards use Flexbox or Grid, as the layout module set them. This section adds the surface: border, padding, and background.

Example

```
/* styles.css */
header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 1rem;
  background-color: #1a365d;
  color: #ffffff;
}

nav a {
  color: #ffffff;
  margin-left: 1rem;
}

.employee-card {
  background-color: #ffffff;
  border: 1px solid #6e7781;
  padding: 1rem;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Nav and cards</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
    <nav>
      <a href="directory.html">Directory</a>
      <a href="leave.html">Leave</a>
    </nav>
  </header>
  <main>
    <article class="employee-card">
      <h2>Ananya Rao</h2>
      <p>Engineering</p>
    </article>
  </main>
</body>
</html>
```

This will produce the following result:



78. Common UI Components

A badge, a button, and a muted note are small classes you reuse. The badge names a status such as Active. Muted text must clear 4.5:1 on the page background. `#3d4654` on `#f4f6f8` clears that bar. The badge text uses the same ink on a light blue chip.

Example

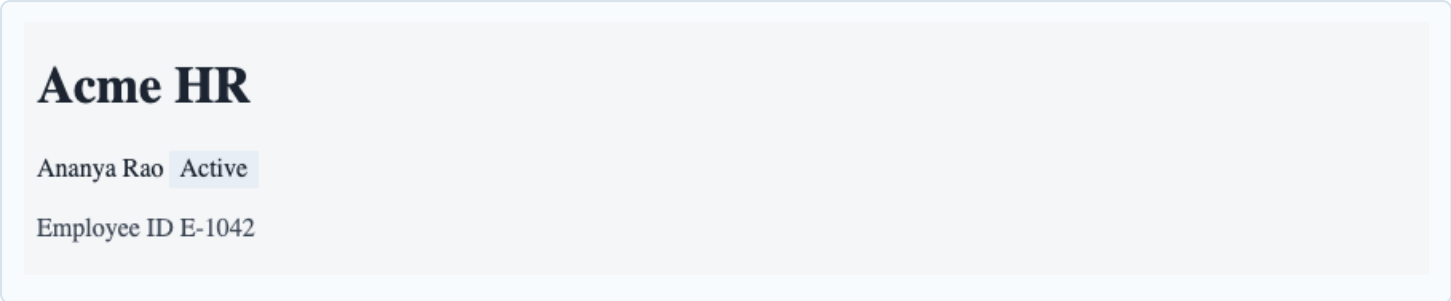
```
/* styles.css */
body {
  background-color: #f4f6f8;
  color: #1b2430;
}

.badge {
  display: inline-block;
  padding: 0.15rem 0.4rem;
  background-color: #e8eef6;
  color: #1b2430;
}

.muted {
  color: #3d4654;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Components</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <p>Ananya Rao <span class="badge">Active</span></p>
    <p class="muted">Employee ID E-1042</p>
  </main>
</body>
</html>
```

This will produce the following result:



79. `:hover`

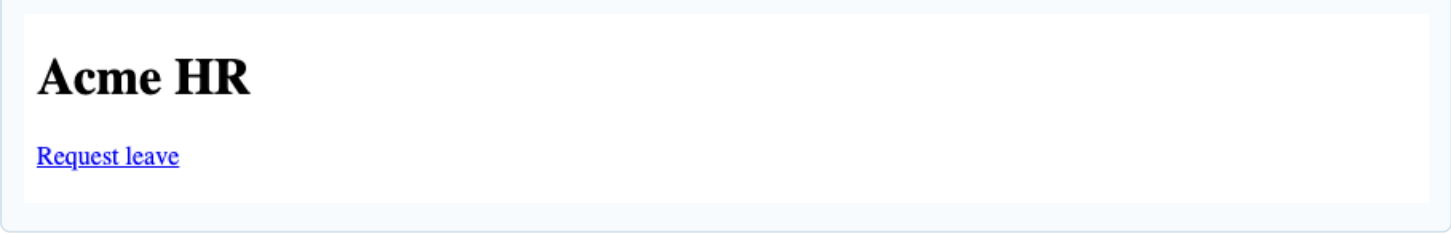
`:hover` matches an element under the pointer. Use it to change a background or an underline. Do not hide information until hover. A phone has no hover. The directory link can darken. The staff name stays visible without it.

Example

```
/* styles.css */
a.button:hover {
  background-color: #0f2744;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Hover</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <a class="button" href="leave.html">Request leave</a>
  </main>
</body>
</html>
```

This will produce the following result:



80. `:focus`

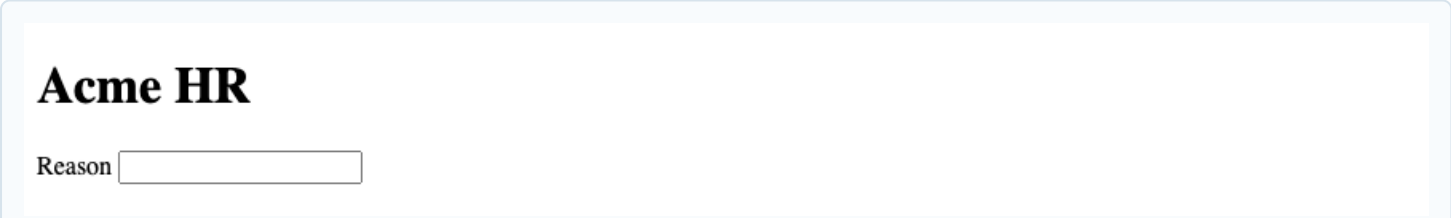
`:focus` matches the element that has keyboard or pointer focus. A field shows focus when the reader tabs to it. Do not remove an outline unless another visible style replaces it. The next section uses `:focus-visible` for the keyboard ring.

Example

```
/* styles.css */
input:focus {
  border-color: #1a365d;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Focus</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <form action="/leave" method="post">
      <p>
        <label for="reason">Reason</label>
        <input id="reason" name="reason" type="text">
      </p>
    </form>
  </main>
</body>
</html>
```

This will produce the following result:



81. `:focus-visible`

Ship `:focus-visible`. It matches when the browser would draw a focus ring for a keyboard user. Do not remove an outline unless another visible style replaces it. On light surfaces use a navy ring. On the navy header that same navy outline disappears, so header links need a light ring.

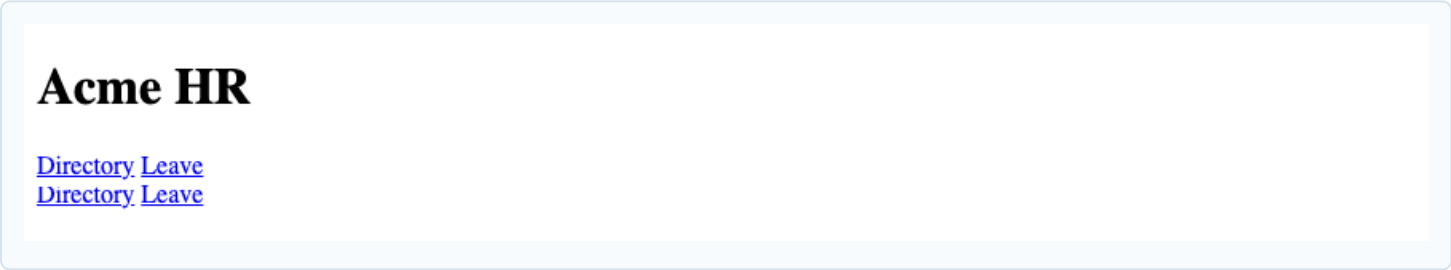
Example

```
/* styles.css */
:focus-visible {
  outline: 2px solid #1a365d;
  outline-offset: 2px;
}

header :focus-visible {
  outline-color: #ffffff;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Focus visible</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
    <nav>
      <a href="directory.html">Directory</a>
      <a href="leave.html">Leave</a>
    </nav>
  </header>
  <main>
    <a href="directory.html">Directory</a>
    <a href="leave.html">Leave</a>
  </main>
</body>
</html>
```

This will produce the following result:



82. `:active`

`:active` matches a control while it is being pressed. A short colour change confirms the press. Keep the label readable. The moment is brief. The focus ring from `:focus-visible` still matters after the press.

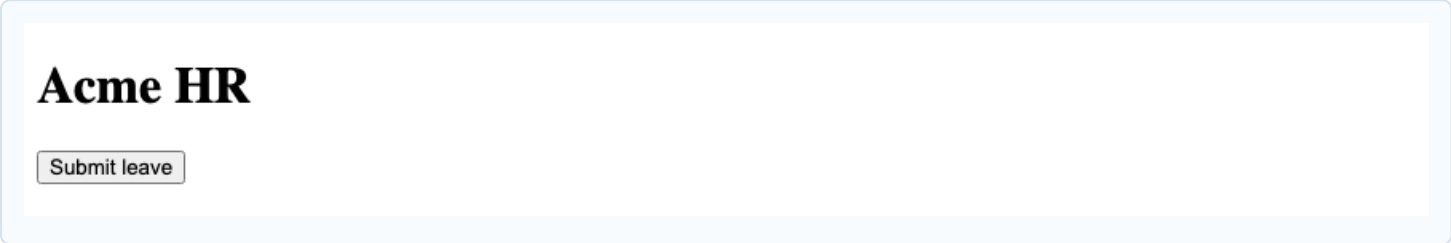
Example

```
/* styles.css */
button:active {
  background-color: #0f2744;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Active</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
```

```
  <main>
    <form action="/leave" method="post">
      <button type="submit">Submit leave</button>
    </form>
  </main>
</body>
</html>
```

This will produce the following result:



83. `:disabled`

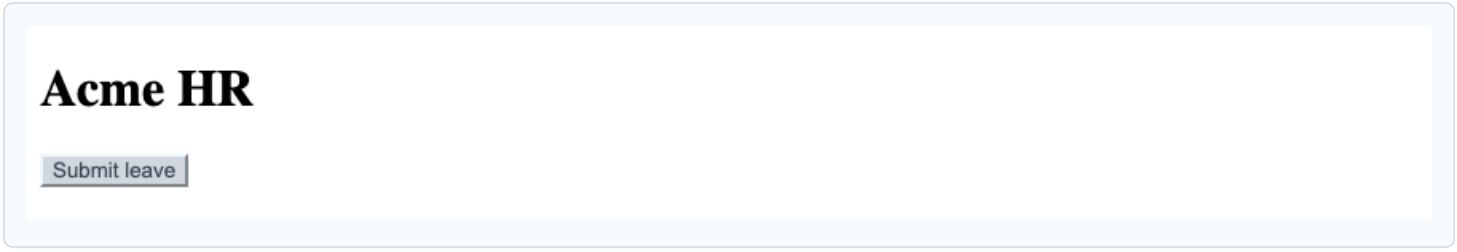
`:disabled` matches a control with the `disabled` attribute. Fade the button and stop the pointer. The HTML still owns the meaning: a disabled control is not submitted. The style only shows that the control is off.

Example

```
/* styles.css */
button:disabled {
  background-color: #d0d7de;
  color: #3d4654;
  border-color: #d0d7de;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Disabled</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <form action="/leave" method="post">
      <button type="submit" disabled>Submit leave</button>
    </form>
  </main>
</body>
</html>
```

This will produce the following result:



84. `:checked`

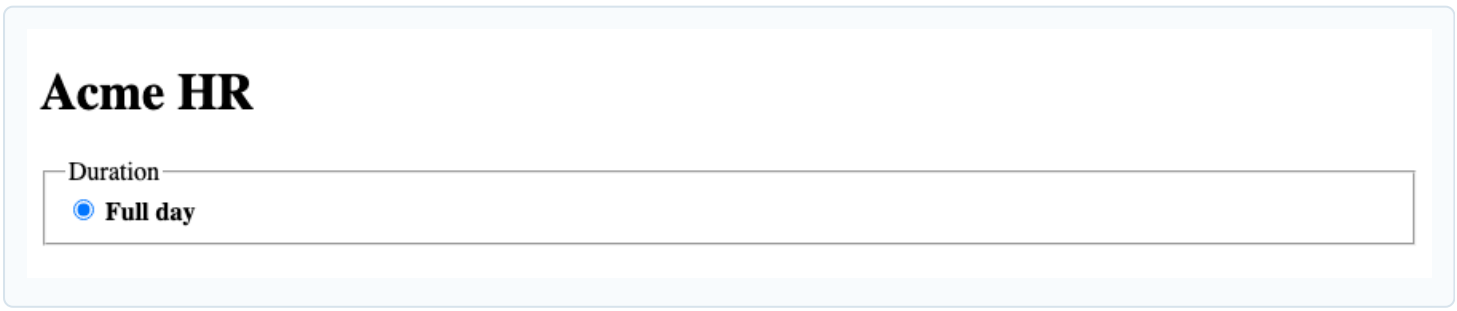
`:checked` matches a selected radio or checkbox. You can style the label that follows a checked box. The control itself stays a real checkbox so the form still submits it. A selected half-day can show a stronger label.

Example

```
/* styles.css */
input:checked + label {
  font-weight: 700;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Checked</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <form action="/leave" method="post">
      <fieldset>
        <legend>Duration</legend>
        <input id="span-full" name="span" type="radio" value="full" checked>
        <label for="span-full">Full day</label>
      </fieldset>
    </form>
  </main>
</body>
</html>
```

This will produce the following result:



85. CSS Transitions

A transition animates a change from one value to another. The button background can take 0.2 seconds. `prefers-reduced-motion` wraps anything that moves. The transition lives inside the `no-preference` query, so a reader who asks for less motion gets an instant change.

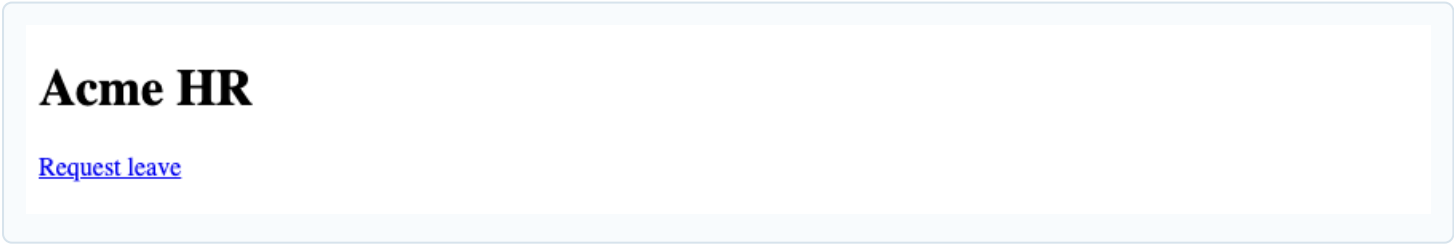
Example

```
/* styles.css */
@media (prefers-reduced-motion: no-preference) {
  a.button {
    transition: background-color 0.2s;
  }
}

a.button:hover {
  background-color: #0f2744;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Transition</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <a class="button" href="leave.html">Request leave</a>
  </main>
</body>
</html>
```

This will produce the following result:



86. CSS Transforms

`transform` moves, scales, or rotates a box without changing the layout around it. A card can lift a few pixels on hover. That move is motion. Pair the transform with a `transition` so the card eases instead of jumping. Wrap both the same way as other motion. The directory does not depend on the lift. The name stays readable if the transform never runs.

Example

```
/* styles.css */
@media (prefers-reduced-motion: no-preference) {
  .employee-card {
    transition: transform 0.2s;
  }

  .employee-card:hover {
    transform: translateY(-2px);
  }
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Transform</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <article class="employee-card">
      <h2>Ananya Rao</h2>
      <p>Engineering</p>
    </article>
  </main>
</body>
</html>
```

This will produce the following result:



87. CSS Animations

An animation runs a set of keyframes, once or in a loop. A transition runs when a property changes (see the button example above). The sample below is an animation: `@keyframes` names the steps, and `animation` applies them. A looping decoration is not part of the Acme HR portal. The labs do not use animations.

Example

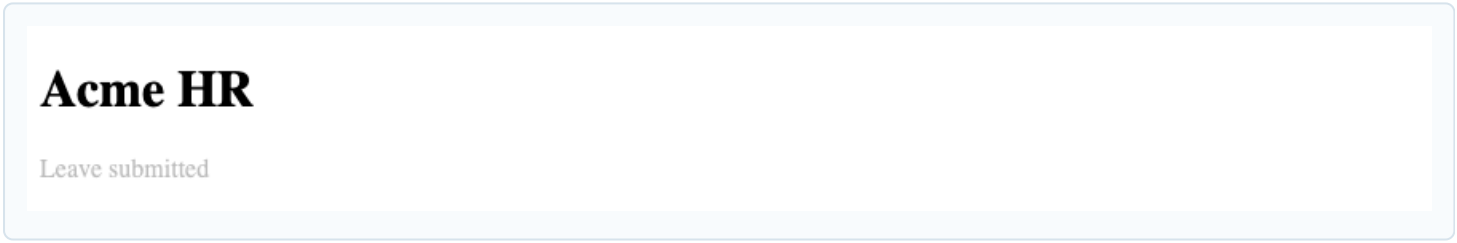
```
/* Demo only — not used on the Acme HR pages or in the labs. */
@keyframes fade-in {
  from { opacity: 0; }
  to { opacity: 1; }
```

```
}

@media (prefers-reduced-motion: no-preference) {
  .status-badge {
    animation: fade-in 0.4s ease-out;
  }
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Animation</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <p class="status-badge">Leave submitted</p>
  </main>
</body>
</html>
```

This will produce the following result:



88. `@keyframes`

`@keyframes` names the steps of an animation. Nothing moves until a rule also sets `animation` (or `animation-name`) that refers to that name. The labs do not use `@keyframes`. Leave looping motion out of the Acme HR pages. The button transition does not need keyframes.

Example

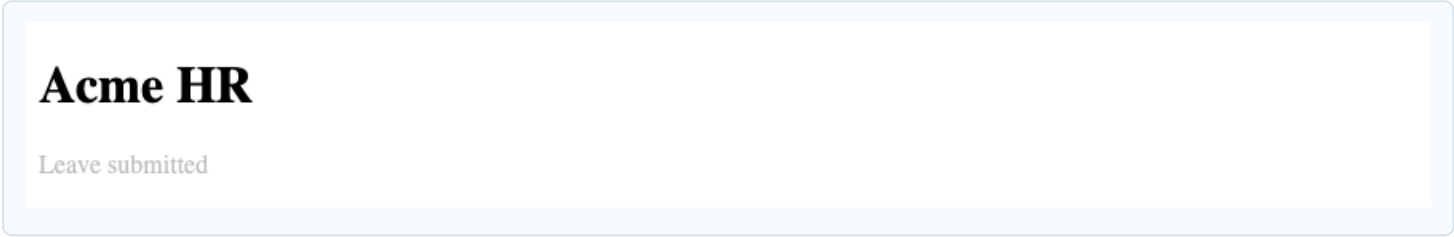
```
/* Demo only — not used on the Acme HR pages or in the labs. */
@keyframes fade-in {
  from { opacity: 0; }
  to { opacity: 1; }
}

.status-badge {
  animation: fade-in 0.4s ease-out;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Keyframes</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <p class="status-badge">Leave submitted</p>
  </main>
</body>
</html>
```

This will produce the following result:



89. Accessibility & Focus States

Ship `:focus-visible`. Do not remove an outline unless another visible style replaces it. Use a light focus ring on the navy header. Muted text must clear 4.5:1 on the page background. Input and card borders need about 3:1 against the adjacent surface. A link needs an underline or another cue besides colour. The payslip amounts stay in a table with headers. The styles do not remove those headers.

Example

```
/* styles.css */
body {
  background-color: #f4f6f8;
  color: #1b2430;
}

.muted {
  color: #3d4654;
}

a {
  color: #1a365d;
}

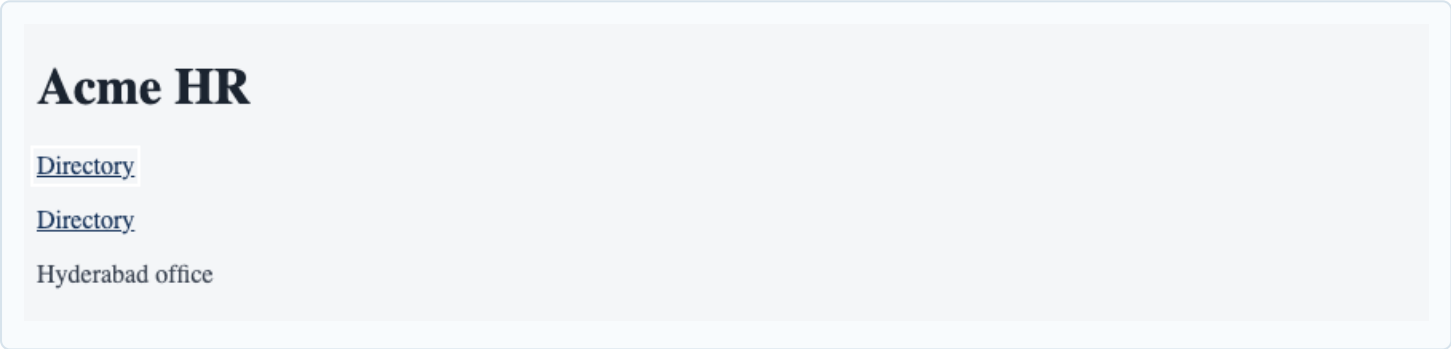
:focus-visible {
  outline: 2px solid #1a365d;
  outline-offset: 2px;
}

header :focus-visible {
  outline-color: #ffffff;
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
```

```
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Focus and contrast</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Acme HR</h1>
    <nav>
      <a href="directory.html">Directory</a>
    </nav>
  </header>
  <main>
    <p><a href="directory.html">Directory</a></p>
    <p class="muted">Hyderabad office</p>
  </main>
</body>
</html>
```

This will produce the following result:



90. Reduced Motion — Basic Understanding

`prefers-reduced-motion` wraps anything that moves. Put transitions and transforms inside `@media (prefers-reduced-motion: no-preference)`. A reader who sets the system to reduce motion then gets an instant colour change and no card lift. The information on the page does not depend on the motion.

Example

```
/* styles.css */
@media (prefers-reduced-motion: no-preference) {
  a.button {
    transition: background-color 0.2s;
  }

  .employee-card:hover {
    transform: translateY(-2px);
  }
}
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Acme HR — Reduced motion</title>
  <link rel="stylesheet" href="styles.css">
</head>
```

```
<body>
  <header>
    <h1>Acme HR</h1>
  </header>
  <main>
    <article class="employee-card">
      <h2>Ananya Rao</h2>
      <p>Engineering</p>
    </article>
    <p><a class="button" href="leave.html">Request leave</a></p>
  </main>
</body>
</html>
```

This will produce the following result:

